

- F. Uplift restraints are undesirable and should only be considered when all other alternatives have been evaluated and only when approved by the SSDE.

Commentary: The use of concrete counterweights in the form of thickened diaphragms or slabs should be investigated before considering uplift restraints.

- G. For segmental box bridges, provide a minimum of two bearings at the end of a unit or at the abutment.

Commentary: Single bearings at the end of a unit or at the abutment are not permitted due to issues with structural stability during construction, and the effects of live load induced torsion and rotation on the expansion joint.

- H. Uplift on all bearing types is undesirable at Service and Strength Limit States, and requires SSDE approval. The erection sequence shown in the plans must provide a statically stable support system that ensures the bearings are in compression for all construction load combinations. Uplift on reinforced elastomeric bearings, without bonded top and/or bottom load plates during construction, shall be limited and approved by the SSDE on a case-by-case basis. For curved segmental bridges constructed in balanced cantilever, temporary bearing uplift may be avoided through the use of counterweights or tie-downs. See also **SDM** 20.9.1.

6.5.1 Design

- A. For bridge bearings specify composite elastomeric bearing pads and other bearing devices that have been designed in accordance with **LRFD** Method B, the **Specifications**, and this document. Specify elastomeric bridge bearing pads by thickness, area, lamination requirement and shear modulus. For normal applications, specify a shear modulus of either 0.110 ksi, 0.130 ksi or 0.150 ksi (at 73 degrees F). For unusual applications, the shear modulus may vary from 0.095 to 0.2 ksi (at 73 degrees F). Do not apply the 1.20 load factor in **LRFD** Table 3.4.1-1 to the thermal movements (TU) for elastomeric bearing pad design when using **LRFD** Method B to determine the total shear deformation in each direction per **LRFD** 14.7.5.3.2. Include the effects of Dynamic Load Allowance for Live Load.

Commentary: Since the calculation of the shear strain in each direction is based on 60% of the total movement, the Department's interpretation of the LRFD Specification is that the 1.2 factor is already built into the empirical equation of LRFD 14.7.5.3.2.

- B. For ancillary structures (noise walls, pedestrian or traffic railings, etc.) use plain elastomeric bearings as typically used on flat-slab bridges and for other applications. Design pads in accordance with **LRFD** Method A, and specify by thickness, area (length and width), and hardness (durometer) or shear modulus (G).
- C. Whenever possible, and after confirming their adequacy, standard designs should be used. See **Standard Plans** Index 400-510 and the associated **Standard Plans Instructions (SPI)**, for standard composite elastomeric bearing pads. Only when the neoprene capacities of the standard pads have been exceeded or when site conditions or constraints dictate provisions for special designs (such as multi-rotational capability) should other bearing systems or components be considered. If other bearing systems or